AOAsNet: A model exploring the correlation of the attention modules is used for medical image classification

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***Abstract*—Medical image classification is of great significance in medical auxiliary diagnosis. The development of deep learning methods has promoted the progress of medical image classification. In current research, most studies use attention modules, but they ignore the in-depth exploration of the relationship between the output of attention modules and the original feature maps. In this study, we first propose a backbone model, then design a framework for integrating multiple attention mechanisms, and finally design a probability distribution-based module to learn the correlation between the output of attention modules and the original feature maps. In the experiments, the AOAsNet model proposed in this study was verified on three public medical datasets. The results show that our method achieves higher accuracy than the benchmarking, and outperforms the comparison models in terms of the number of model parameters, FLOPs, and training time cost.**

***Index Terms*—Medical Image Classification , Attention, Feature Association.**

I. Introduction

Deep learning, as a pivotal branch within the expansive realm of machine learning, has demonstrated immense potential in the field of medical image analysis. This potential stems from its remarkable capabilities in feature extraction and nonlinear mapping, which enable it to discern intricate patterns and relationships within medical images that may elude human observation. At this pivotal stage of research, a diverse array of models has emerged, each tailored to address specific challenges in medical image processing. These models have been successfully applied to a wide range of tasks, including multi-center data alignment, brain glioma detection, skin cancer classification and segmentation, among others [1] [2] [3] [4] [5] [6].

As the intersection between medical image processing and deep learning deepens, a cutting-edge artificial intelligence technology—the attention module—is gradually infiltrating and profoundly reshaping the landscape of medical image analysis. The attention module, inspired by the human visual system's ability to focus on relevant information while ignoring irrelevant details, has proven to be a powerful tool in enhancing the performance of deep learning models. The integration of this innovative technology into medical image analysis not only significantly improves the accuracy and efficiency of medical image diagnosis but also opens up a plethora of new possibilities for clinical decision support, thereby facilitating more personalized and precise medical treatments.

In the current research landscape, numerous scholars have taken inspiration from the successful application of attention modules in the field of natural images and adapted them for use in medical images. Among these adaptations are the integrated spatial-frequency attention module, which captures both spatial and frequency domain information; grid-space attention, which focuses on local regions within the image grid; and graph attention, which leverages graph structures to model relationships between image elements [7] [8] [9].

Building upon these advancements, in this study, we have designed a comprehensive framework for integrating multiple attention modules. Specifically, we have incorporated the CBAM [10], SGCA [11], and GLCM [12] attention modules, all of which have demonstrated exceptional performance in previous research. Furthermore, to enhance the framework's

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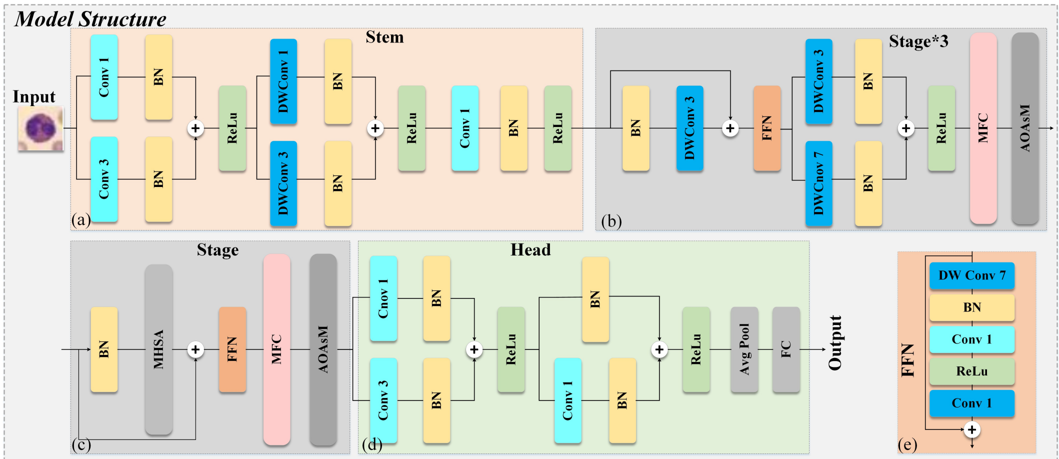

Fig. 1. Model structure.

ability to extract and merge features from multiple perspectives, we have designed a novel LA attention module based on the Laplace transform. This innovative approach leverages the mathematical properties of the Laplace transform to capture and emphasize key features within medical images, thereby improving the overall diagnostic accuracy and robustness of the model.

In this study, we have made the following contributions: 1. We designed and proposed a network model combining CNN and ViT architectures for medical image classification. 2. We presented a framework for integrating and merging multiple attention modules. 3. We developed a correlation acquisition module for the original feature maps and the feature maps of attention modules to explore the potential correlations among the model's feature maps. 4. We conducted experiments on the BloodMNIST (28×28), DermaMNIST (28×28), and Pneumonia (224×224) datasets, and compared our model with 14 competing algorithms from the literature. The code can be obtained on github: https://github.com/Zhengzechenzzc/AOAsNet

II. Methods

A. Model Structure

This study design a network architecture with 1 Stem, 4 Stages, and 1 Head. Each Stage halves resolution and doubles channels to capture multi-scale features. The first three Stages share a layout of depth-separable 3×3 Convolutions (DWConv) and feedforward neural networks (FFN), enhancing feature extraction while reducing parameters. The final Stage innovatively integrates multi-head self-attention for global context perception. Each Stage ends with a Multi-Feature Concat Module (MFC) for multi-scale fusion and a Attention and Original feature Association Module (AOAsM) for feature optimization. The Stem employs Conv 3×3+DWConv 3×3+Conv 1×1 for initial feature extraction. The Head uses Conv 3×3+Conv 1×1+Avgpool+FC for final output, enabling feature integration and classification. FFN balances efficiency and performance with DWConv 7×7+Conv 1×1×2, and adds residual connections for stability and performance.

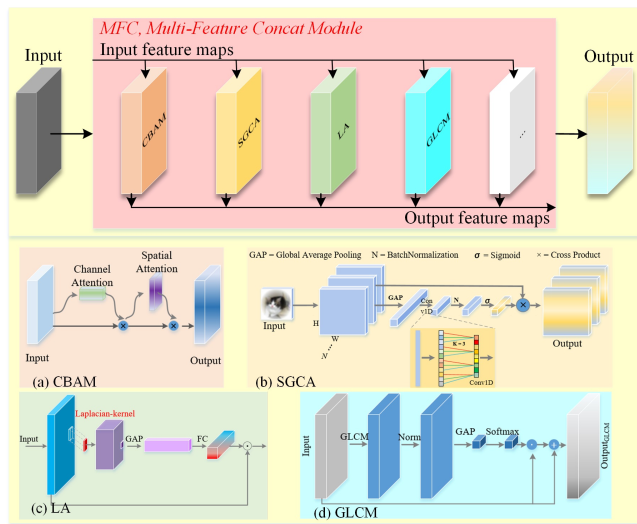

Fig. 2. Multi-Feature Concat Module.

B. Multi-Feature concat module

Studies show attention mechanisms enhance deep learning in medical images. We designed an integrated framework combining multiple attention modules (CBAM, SGCA, LA, GLCM) in the Multi-Feature Concat Module (MFC). MFC inputs $x \in P^{B \times C \times H \times W}$,and outputs a tensor of the same shape. Each module's weight w_i starts equal and adjusts during training.

$$w = w_1, w_2, ..., w_N, w_i \in P^{+} \tag{1}$$

As in formula (2), the output of the selected attention module is weighted and fused,

$$y = \sum_{i \in S} w_i \cdot AttentionModule_i(x) \tag{2}$$

Where $S \subseteq 0, 1, ..., N-1$ is the selected collection of attention modules, $AttentionModule_i(x)$ is the output of the

i attention module, w_i is the weight of the corresponding module.

In the MFC module, a Laplacian Attention (LA) based attention module is designed. The edge and high frequency information are extracted by Laplace transform of the input feature map.

$$Laplacian(x) = x * K_{Laplacian} \tag{3}$$

Where $x \in P^{B\times C\times H\times W}$ is the input feature map, $K_{laplacian}$ is the Laplacian convolution kernel of 3 × 3.

$$K_{laplacian} = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix} \tag{4}$$

After the feature map passes through the Laplace convolution kernel, the channel level features are extracted using the global average pooling, and the channel level features are obtained (5).

$$P_c = \frac{1}{H \times W} \sum_{h=1}^{H} \sum_{w=1}^{W} L_{c,h,w} \tag{5}$$

Where $L_{c,h,w}$ is the output of the Laplace transform of channel c . The channel weights are then generated using two fully connected layers.

$$w = \sigma(W_2 ReLu(W_1 p)) \tag{6}$$

Where W_1 and W_2 is a linear parameter, σ is sigmoid. Finally, the original feature graph x is weighted.

$$y_c = x_c \cdot (1 + w_c) \tag{7}$$

Where w_c is the weight of the c channel.

C. Attention and Original feature Association Module

In classification tasks, especially when multiple attention mechanisms are used simultaneously, it is crucial to effectively capture the correlations between these attention modules and the original features. By analyzing their correlations, we can identify and quantify the mutual influence effect of the feature maps output by the attention mechanism on the original features, which is of crucial significance for model optimization and understanding learning tasks. To this end, we propose a module (Attention and Original feature Association Module (AOAsM)), which is specifically used to study the framework of the correlation relationship among the outputs of different attention modules. This module enables the model to automatically learn how to perform weighted fusion on the output of each attention module based on the output of other modules.

First, the input original feature maps ($X_{original}$) and attention feature maps ($X_{attention}$) are encoded through VAE to generate potential representations $z_{original}$ and $z_{attention}$.

$$z_{original} = Encoder(x_{original}) \tag{8}$$

$$z_{attention} = Encoder(x_{attention}) \tag{9}$$

Then calculate their differences:

$$\Delta z = z_{original} - z_{attention} \tag{10}$$

The potential representation difference Δz is then calculated and its probability density function is estimated using the Student's distribution (PDF). The probability density function of the Student't distribution is (11).

$$f(s|v, \mu, \sigma) = \frac{\Gamma(\frac{v+1}{2})}{\sqrt{v \cdot \pi} \cdot \sigma \cdot \Gamma(\frac{v}{2})} (1 + \frac{(x-\mu)^2}{v\sigma^2})^{-\frac{v+1}{2}} \tag{11}$$

Where $v > 0$ is the degree of freedom; μ is the position parameter; $\sigma > 0$ indicates the scale parameter.

A neural network is used to learn the parameters of the Student't distribution: degrees of freedom *v*, position μ, and scale σ .

$$\mu, \sigma = softplus(Linear(\Delta z)) \tag{12}$$

$$v = softplus(v_{unconstrained}) + 1 \tag{13}$$

$softplus(x) = log(1 + e^x)$, ensure $v > 0, \sigma > 0$. Model Δz using the PDF of the Student't distribution to calculate the association weight:

$$w = exp(log\ p(\Delta|v, \mu, \sigma)) \tag{14}$$

Expand the PDF to:

$$\begin{aligned} log\ p(\Delta z|v, \mu, \sigma) = -\frac{v+1}{2} log(1 + \frac{(\Delta z - \mu)^2}{v\sigma^2}) - \\ (log\ \sigma + \frac{1}{2} log\ v + \frac{1}{2} log\ \pi + log\ \Gamma(\frac{v}{2}) - log\ \Gamma(\frac{v+1}{2})) \end{aligned} \tag{15}$$

$$y = (\Delta z - \mu)/\sigma \tag{16}$$

$$z = log\ \sigma + \frac{1}{2} log\ v + \frac{1}{2} log\ \pi + log\ \Gamma(\frac{v}{2}) + log\ \Gamma(\frac{v+1}{2}) \tag{17}$$

y is the normalized difference, and z is the constant term, which is also the normalized factor, including the logarithm of scale and degrees of freedom. And the weight is (18),

$$w = exp(-\frac{v+1}{2} log\ (1 + \frac{y^2}{v}) - Z) \tag{18}$$

Next, the potential representation of the attention feature map is generated by the decoder to reconstruct the feature map.

$$\hat{x}_{attention} = Decoder(z_{attention}) \tag{19}$$

Finally, the association weight w is used to fuse the original feature map and the reconstructed feature map.

$$x_{concat} = w \cdot \hat{x}_{attention} + (1 - w) \cdot x_{original} \tag{20}$$

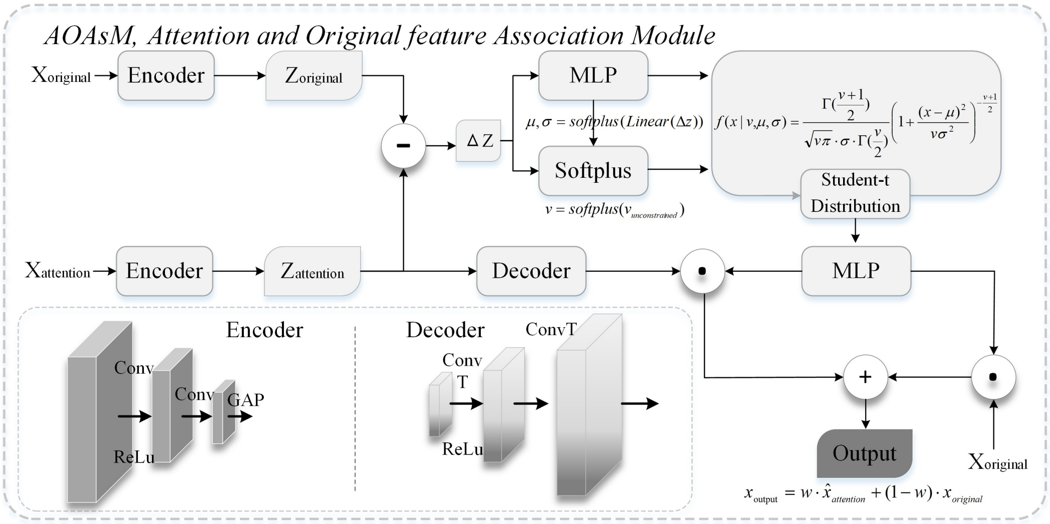

Fig. 3. Attention and Original feature Association Module(AOAsM).

TABLE I
RESULTS OF AOASNET AND OTHERS ON THE BLOODMNIST AND DERMAMNIST.

Model	BloodMNIST		DermaMNIST		Params(M)	FLOPs(M)
	Acc	Time Cost	Acc	Time Cost		
ResNet18(224) Benchmarking	**96.30%**	-	**75.40%**	-	-	-
MobileNetv3	95.44%	75.3m	70.37%	177.2m	4.21	233.57
GoogleNet	96.20%	126.3m	74.56%	38.7m	9.95	1510.32
ConvNext	91.38%	1128m	68.08%	598.7m	87.57	15372.43
ResNext	94.74%	130m	74.76%	25.4m	23.00	87.49
ViT	**96.93%**	670m	**76.81%**	778.7m	85.80	11285.49
Swin-Transformer	94.53%	404m	74.81%	411.8m	86.75	10222.11
Efficientnet	70.56%	335.1m	66.88%	392.7m	52.87	5445.22
GhostNet	90.68%	**17.7m**	71.57%	**23.6m**	39.12	194.87
EfficientViT	91.32%	508.9m	67.53%	234.5m	21.58	**79.92**
EdgeNeXt	88.95%	124.9m	72.17%	201.8m	**1.79**	283.59
AOAsNet(Ours)	**96.60%**	**159m**	**77.61%**	**184.5m**	**11.06**	**526.88**

III. EXPERIMENT

A. Experiment settings

Datasets:Three datasets containing two small scale medical public datasets and one large scale medical public dataset, BloodMNIST, DermaMNIST, Pneumonia(from MedMNIST [12]), were used in the experiment. We used some common enhancement techniques, namely random resizing and cropping.

Experiment settings:The input to the data is the pixel size of the dataset, including 28× 28 and 224 × 224. The optimizer selects stochastic gradient Descent (SGD) to optimize the network parameters, with weight attenuation of 1e-4, momentum of 0.9,batchsize of 32, initial learning rate of 0.1, and model training of 200 epochs with half of the original attenuation every 10 epochs. The experimental environment is pytorch1.12 version, python = 3.9, hardware using NVIDIA RTX3060 12G * 2, memory 64G., i7-13700KF.

B. Comparative experiments

In the experiment, we used parametes, FLOPs and training time cost respectively to demonstrate the performance of the model, and used accuracy to verify the effectiveness of the model. The results are shown in Table 1 and Table 2. The AOAsNet parameter proposed in this study is 11.06M, which is equivalent to 1/8 of the parameters of Swin-Transformer, which is relatively lightweight in many models and greatly reduces the parameter compared with other VIT models.

On BloodMNIST and DermaMNIST datasets, AOAsNet's FLOPs are 526.88M and 526.87M respectively, which is equivalent to 1/29 of ConvNexT's, and compared with 2.25 times of MobileNetv3, the overall computational complexity is low. In terms of training time cost, AOAsNet spent 159 minutes and 184.5 minutes on the BloodMNIST and DermaMNIST datasets, respectively, with an overall faster rate. AOAsNet achieved 96.60% accuracy on the BloodMNIST dataset, which is more accurate than benchmarking, 0.33% lower than ViT, but significantly better in terms of reference count, FLOPs, and time cost. The accuracy of 77.61% was

TABLE II
RESULTS OF AOASNET AND OTHERS ON THE PNEUMONIA.

Model	Acc	Params(M)	FLOPs(M)	Time Cost
ResNet50(224) benchmarking	**88.40%**	-	-	-
MobileNetv3	90.22%	4.21	233.56	132.5m
GoogleNet	91.51%	9.95	1510.31	62m
ConvNext	87.34%	87.57	15375.34	343.3m
ResNext	91.67%	23.00	87.48	120.1m
ViT	89.26%	85.80	11285.49	375.6m
Swin-Transformer	88.78%	86.75	10222.11	468.1m
Efficientnet	85.90%	52.87	5445.21	378.4m
GhostNet	**92.31%**	39.12	194.86	**38.1m**
EfficientViT	84.29%	21.58	**79.91**	92.1m
EdgeNeXt	86.86%	**1.79**	283.59	140.6m
AOAsNet(ours)	**92.95%**	**11.06**	**25156.77**	**348.2m**

achieved on the DermaMNIST dataset, which is 0.80% higher than ViT and 2.21% higher than benchmarking.

In the Pneumonia dataset, the FLOPs of AOAsNet was 25,156.77m and the time cost was 348.2 minutes. Compared with the small scale datasets, the computational complexity and time cost of AOAsNet was greatly increased, but the accuracy rate was 92.95%. The accuracy rate is far higher than 88.40% for benchmarking and 0.64% for GhostNet.

The experimental results of AOAsNet and the comparison methods on the three datasets generally demonstrate the performance of our proposed method, and generally obtain a relatively obvious advantage.

C. Ablation studys

This section presents ablation studies for AOAsNet. We added MFC and AOAsM to the backbone network (Section 2.1) to test performance. Table 3 shows experiments on three datasets. Adding MFC increased parameters by 0.2M and slightly raised computational complexity. Adding AOAsM increased parameters by 1.04M and computational complexity by 34-14 times. Despite AOAsM significantly raised computational complexity, but the parameter increment being small and time cost only increased by 17.86%-25.59% with AOAsM. Model accuracy improved by 0.25%-1.39% with MFC and by 0.52%-2.04% with AOAsM. The AOAsM module significantly enhanced the model, proving the importance of the association between features in feature map transmission.

TABLE III
ABLATION STUDYS.

Datasets	Model	Acc	Params(M)	FLOPs(M)	Time Cost
BloodMNIST	backbone	95.83%	9.82	34.73	127.4m
	+MFC	96.08%	10.02	35.02	130.6m
	+MFC+AOAsM	96.60%	11.06	526.88	159m
DermaMNIST	backbone	74.12%	9.82	34.73	144.3m
	+MFC	75.57%	10.02	35.02	148.4m
	+MFC+AOAsM	77.61%	11.06	526.87	184.5m
Pneumonia	backbone	90.28%	9.82	1696.77	252.5m
	+MFC	91.67%	10.02	1697.71	259.1m
	+MFC+AOAsM	92.95%	11.06	25156.77	348.2m

D. Visualized Analysis

In Figure 4, a compelling visual narrative is presented regarding the impact of heatmap mapping based on the attention module's correlation on the interpretability of deep learning models in medical and biological image analysis. The figure shows the original images from the BloodMNIST and DermaMNIST datasets and their corresponding attention maps visualized after processing with AOAsM and MFC methods. A careful observation reveals that the hotspots in the heatmaps generated by AOAsM are more precisely concentrated around the key feature points in the original images. For instance, in the BloodMNIST images depicting blood cells, the attention maps generated by the MFC method appear more scattered and less focused, while those generated by AOAsM clearly and precisely highlight the cell structures and internal components. AOAsM further refines the relationships of features on the basis of MFC. Similarly, in the DermaMNIST images showing skin lesions, the AOAsM method successfully locates the key areas of the lesions, such as the edges and color changes.

The concentrated heat distribution in the AOAsM heatmap strongly indicates that during the training process, AOAsM can capture more associations between input features and model predictions. Unlike traditional attention mechanisms that may only highlight significant regions but lack in-depth understanding of underlying relationships, AOAsM delves deeper into the data to identify the true drivers of model decisions. By leveraging these associations, AOAsM effectively focuses attention on the most relevant features in the image, thereby providing a more accurate and interpretable representation of the model's decision-making process. In summary, the visual evidence presented in Figure 4 highlights the superiority of the AOAsM method in revealing the relationship between the attention module and the original image in complex image data, and utilizes these relationships to generate more concentrated and meaningful attention maps, which is crucial for enhancing the interpretability and credibility of deep learning models in critical applications such as medical diagnosis.

IV. CONCLUSION

In this study, a AOAsNet is proposed for medical image classification. The experimental results show that the backbone, MFC and AOAsM modules contained in AOAsNet have better results in the classification process, and the model has lower parameters and training time cost. In the future, we will focus on the reduction of computational complexity of AOAsM modules, and integrate the association between features into more complex tasks to improve the ability of medical image classification models.

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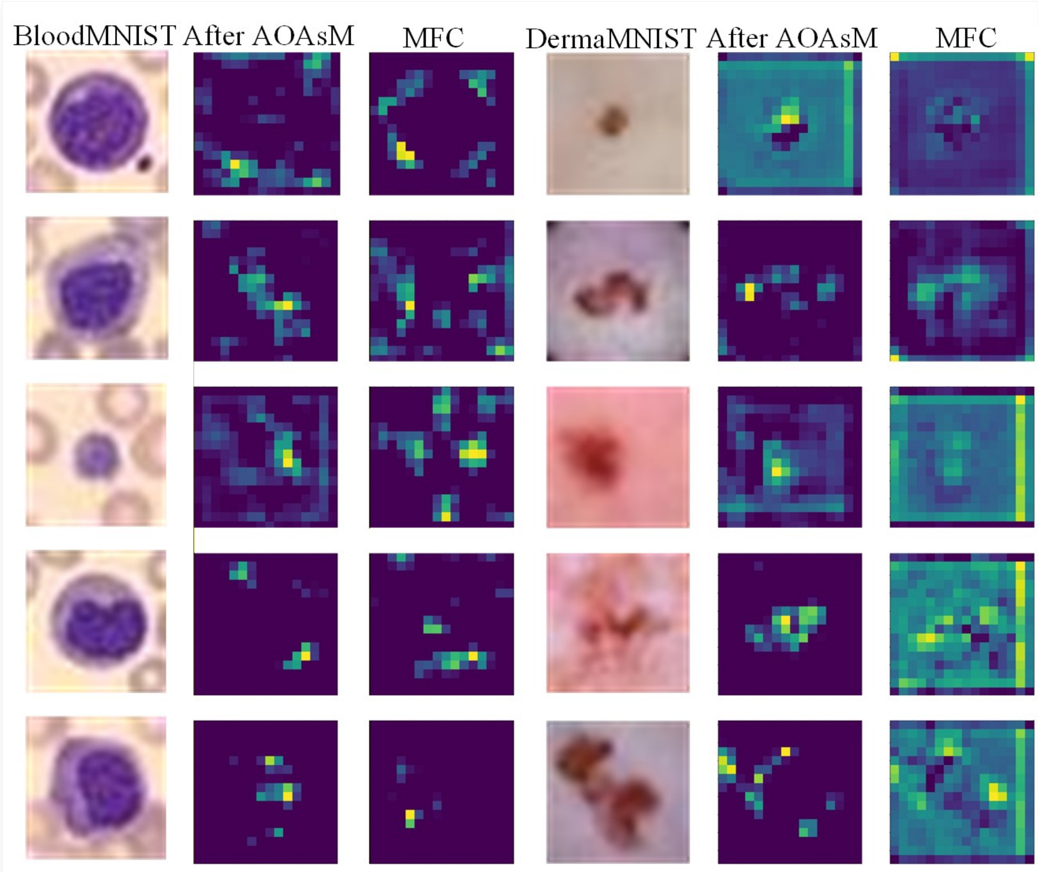

Fig. 4. Visual comparison of hotmaps of data after MFC and AOAsM.

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